WORKED SOLUTION

STEP III 2008, Question 3

Pure • Ellipse Tangents and Foci

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Read the question before you go on. It is in the STEP Support Programme's question database:

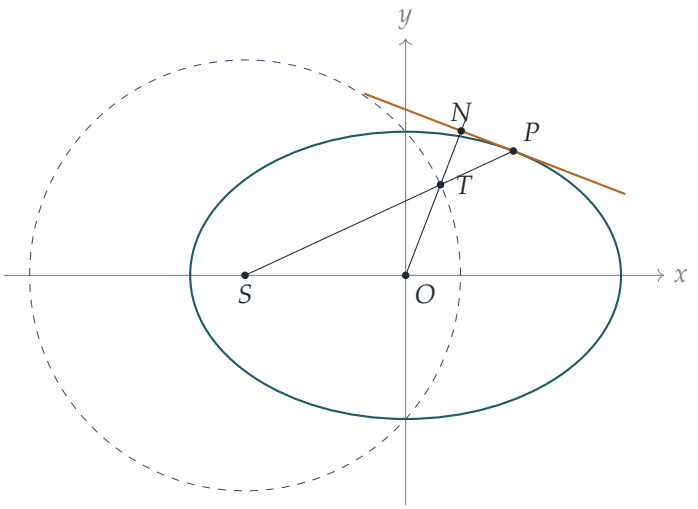
step.maths.org/questions/08-s3-q3

The key idea

Everything here is about two straight lines, so the job is to write down their equations cleanly and intersect them. The line ON is perpendicular to the tangent, so its direction is the tangent's normal vector, which we can read straight off the tangent's equation. Parametrising SP from S makes the intersection a one-line calculation, and the parameter we find turns out to be exactly what is needed for the circle: it tells us what fraction of the way along SP the point T sits.

Solution

Throughout, $0 < e < 1$: we have $e^2 = 1 - b^2/a^2$, which lies strictly between 0 and 1 because $a > b > 0$. We also take $\sin\theta \neq 0$. If $\sin\theta = 0$ then $P = (\pm a, 0)$, the tangent is $x = \pm a$, $N = P$, and both ON and SP are the x-axis, so T isn't defined.



(The picture has $a = 3$, $b = 2$, $\theta = 60^\circ$. The ellipse is teal, the tangent at P amber, and the dashed circle has centre S and radius a.)

The y-coordinate of T

Differentiating $x^2/a^2 + y^2/b^2 = 1$ implicitly, or using the standard result, the tangent at (x_1, y_1) is $xx_1/a^2 + yy_1/b^2 = 1$. At P this is

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1.$$

A line $px + qy = 1$ has normal vector (p, q), so the tangent has normal $(\frac{\cos\theta}{a}, \frac{\sin\theta}{b})$, or, multiplying by ab, $(b\cos\theta, a\sin\theta)$. The line ON passes through O in this direction (it is perpendicular to the tangent), so its points are $\mu(b\cos\theta, a\sin\theta)$, and its equation is

$$a\sin\theta\, x = b\cos\theta\, y. \tag{1}$$

Now parametrise SP from S: its points are

$$(-ea + \lambda a(\cos\theta + e),\ \lambda b\sin\theta),$$

with $\lambda = 0$ at S and $\lambda = 1$ at P. Substituting into (1) and cancelling the factor $\sin\theta \neq 0$,

$$a(-ea + \lambda a(\cos\theta + e)) = \lambda b^2\cos\theta$$
$$\lambda(a^2\cos\theta + a^2e - b^2\cos\theta) = ea^2.$$

Here is where the focus condition earns its keep: $a^2 - b^2 = a^2e^2$, so the bracket is $a^2e^2\cos\theta + a^2e = a^2e(1 + e\cos\theta)$. Dividing by $a^2e \neq 0$,

$$\lambda = \frac{1}{1 + e\cos\theta},$$

which is legitimate since $1 + e\cos\theta \geq 1 - e > 0$. The y-coordinate of T is therefore

$$y_T = \lambda b\sin\theta = \frac{b\sin\theta}{1 + e\cos\theta},$$

as required. (For later reference, (1) then gives $x_T = \dfrac{b\cos\theta}{a\sin\theta}\, y_T = \dfrac{a(1 - e^2)\cos\theta}{1 + e\cos\theta}$.)

T lies on the circle centre S, radius a

Ask yourself We know what fraction of SP the segment ST is. What else would we need?

Since $T = S + \lambda(P - S)$ with $\lambda > 0$, we have $ST = \lambda\, SP$. So we only need the length SP, the focal distance of P:

$$\begin{aligned} SP^2 &= (a\cos\theta + ea)^2 + b^2\sin^2\theta \\ &= a^2[\cos^2\theta + 2e\cos\theta + e^2 + (1 - e^2)\sin^2\theta] \\ &= a^2[(\cos^2\theta + \sin^2\theta) + 2e\cos\theta + e^2(1 - \sin^2\theta)] \\ &= a^2[1 + 2e\cos\theta + e^2\cos^2\theta] = a^2(1 + e\cos\theta)^2, \end{aligned}$$

using $b^2 = a^2(1-e^2)$ in the second line. As $a > 0$ and $1 + e\cos\theta > 0$, this gives $SP = a(1 + e\cos\theta)$. Hence

$$ST = \lambda\, SP = \frac{a(1+e\cos\theta)}{1+e\cos\theta} = a,$$

so T lies on the circle with centre S and radius a.

(If you'd rather work directly from coordinates: $x_T + ea = \dfrac{a(\cos\theta + e)}{1 + e\cos\theta}$, and $(x_T + ea)^2 + y_T^2$ reduces to a^2 by exactly the same algebra as SP^2 above.)

Checks

- At $\theta = \frac{\pi}{2}$, $P = (0, b)$, the tangent is $y = b$, so ON is the y-axis and $T = (0, y_T)$ with $y_T = b$ from the formula. Then $ST^2 = e^2a^2 + b^2 = a^2$, as it should be.
- With $a = 3$, $b = 2$, $\theta = 60^\circ$ (the picture), intersecting the two lines numerically gives $T \approx (0.4857, 1.2618)$; the formula gives $y_T = \sqrt{3}/(1 + \sqrt{5}/6) \approx 1.2618$, and $ST \approx 3.0000$.
- In the circle limit $e \to 0$, $S \to O$ and $T \to P$, which does lie on the circle of radius a about O.

Common slips

- Taking ON parallel to the tangent rather than perpendicular to it: the direction of ON is the normal $(b\cos\theta, a\sin\theta)$, not $(-a\sin\theta, b\cos\theta)$.
- Forgetting to replace $a^2 - b^2$ by a^2e^2, which leaves an expression for λ that doesn't simplify.
- Dividing by $\sin\theta$ without comment; the case $\sin\theta = 0$ genuinely fails.

Beyond STEP

The result $SP = a(1 + e\cos\theta) = a + ex_P$ is the focus–directrix property in disguise: distance to the focus equals e times distance to the directrix $x = -a/e$. Writing a conic in polar coordinates about a focus, $r = \ell/(1 + e\cos\phi)$, is how Kepler orbits are described in first-year mechanics, and the same denominator $1 + e\cos\theta$ appeared here. For the geometry of conics, and many more STEP-style problems, see Siklos, *Advanced Problems in Mathematics*; for orbits, Kleppner and Kolenkow, *An Introduction to Mechanics*.

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